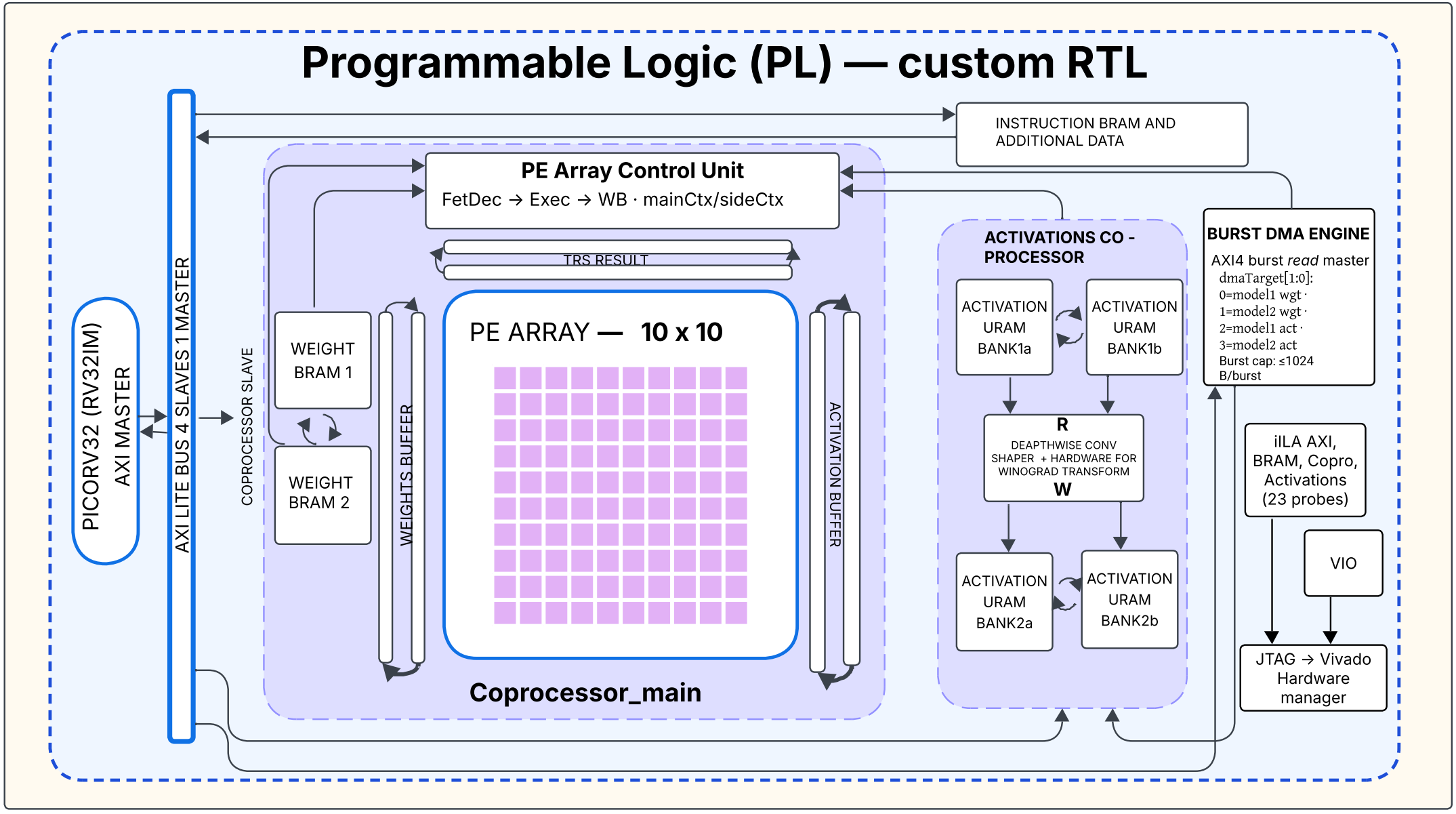

Programmable Logic (PL) — custom RTL
PICORV32 (RV32IM)
AXI MASTER
AXI LITE BUS 4 SLAVES 1 MASTER
COPROCESSOR SLAVE
INSTRUCTION BRAM AND ADDITIONAL DATA
PE Array Control Unit
FetDec → Exec → WB · mainCtx/sideCtx
TRS RESULT
WEIGHT BRAM 1
WEIGHT BRAM 2
WEIGHTS BUFFER
PE ARRAY — 10 x 10
ACTIVATION BUFFER
Coprocessor_main
ACTIVATIONS CO - PROCESSOR
ACTIVATION URAM BANK1a
ACTIVATION URAM BANK1b
R
DEAPTHWISE CONV SHAPER + HARDWARE FOR WINOGRAD TRANSFORM
W
ACTIVATION URAM BANK2a
ACTIVATION URAM BANK2b
BURST DMA ENGINE
AXI4 burst read master
dmaTarget[1:0]:
0=model1 wgt ·
1=model2 wgt ·
2=model1 act ·
3=model2 act
Burst cap: ≤1024 B/burst
iILA AXI, BRAM, Copro, Activations (23 probes)
VIO
JTAG → Vivado Hardware manager